

OBARSIA LOTRULUI, Romania

WMO: 152970

Lat: 45.4356N

Lon: 23.6308E

Elev: 1350

StdP: 86.12

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.4	(c) -16.6	(d) -23.2	(e) 0.5	(f) -18.2	(g) -20.1	(h) 0.7	(i) -16.3	(j) 5.3	(k) -4.7	(l) 4.4	(m) -6.0	(n) 1.3	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 24.0	(d) 15.2	(e) 22.3	(f) 14.8	(g) 20.8	(h) 13.9	(i) 16.2	(j) 22.0	(k) 15.4	(l) 21.0	(m) 14.7	(n) 19.9	(o) 1.3	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	11.8	18.1	13.2	11.2	17.4	12.4	10.6	16.9	50.4	22.3	47.9	21.1	45.8	20.2	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.1	(b) 3.4	(c) 3.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 4.1	(b) 3.4	(c) 3.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-5.8	-3.1	-0.6	3.1	8.4	12.2	13.3	13.7	9.2	5.4	0.7	-3.7	(6)
(7)		DBStd	7.60	5.04	4.21	3.82	4.40	2.66	2.91	2.60	2.36	3.41	3.14	3.93	4.14	(7)
(8)		HDD10.0	2379	489	368	328	209	63	12	5	3	56	142	278	426	(8)
(9)		HDD18.3	5074	747	602	586	458	309	184	157	144	274	400	528	684	(9)
(10)		CDD10.0	348	0	0	0	1	12	78	107	118	32	1	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH23.3	82	0	0	0	0	2	15	30	35	1	0	0	0	(12)
(13)		CDH26.7	6	0	0	0	0	1	0	4	2	0	0	0	0	(13)
(14)	Wind	WSAvg	1.1	1.3	1.2	1.1	1.0	1.0	0.9	1.0	0.9	1.2	0.9	1.5	(14)	
(15)	Precipitation	PrecAvg	815	39	33	46	67	93	114	117	80	75	58	52	51	(15)
(16)		PrecMax	1014	79	92	100	113	193	220	263	134	150	118	187	110	(16)
(17)		PrecMin	534	9	7	11	17	37	28	44	21	9	1	1	8	(17)
(18)		PrecStd	142	21	20	24	26	37	44	56	30	38	32	38	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	9.2	13.5	18.9	21.5	25.0	26.2	26.2	22.2	19.8	12.5	5.1	(19)
(20)			MCWB	2.5	5.3	6.8	9.5	12.6	15.0	15.5	16.0	13.1	9.8	7.3	1.8	(20)
(21)		2%	DB	3.6	6.7	10.4	16.6	18.8	23.1	23.6	23.9	19.7	17.3	10.1	3.2	(21)
(22)			MCWB	1.5	2.5	4.5	9.0	11.5	14.8	15.5	15.1	12.7	9.4	6.6	1.0	(22)
(23)		5%	DB	2.1	4.9	7.5	14.1	16.6	21.1	21.9	22.0	18.0	14.7	8.3	2.1	(23)
(24)			MCWB	0.4	2.1	3.4	7.6	10.7	14.1	14.9	14.8	12.1	8.9	5.8	0.5	(24)
(25)		10%	DB	0.7	3.1	5.1	11.0	14.6	19.2	20.2	20.3	15.9	11.7	6.6	1.2	(25)
(26)			MCWB	-0.6	1.1	2.0	6.4	9.8	13.2	14.0	14.2	11.7	7.6	4.7	0.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	5.6	7.2	10.6	13.4	17.0	17.1	17.4	15.0	11.2	8.2	2.7	(27)
(28)			MCDWB	4.9	8.6	12.9	17.3	19.6	22.1	22.4	22.7	20.2	16.3	11.6	4.3	(28)
(29)		2%	WB	1.8	3.5	5.0	9.4	12.2	15.4	16.0	16.3	13.7	10.2	7.2	1.5	(29)
(30)			MCDWB	3.4	5.7	9.4	16.1	17.3	21.8	21.9	22.1	18.1	15.4	9.7	2.7	(30)
(31)		5%	WB	0.6	2.3	3.6	8.2	11.2	14.5	15.2	15.4	12.8	9.3	6.2	0.8	(31)
(32)			MCDWB	1.8	4.4	7.0	13.7	15.9	20.6	20.6	20.9	17.0	14.3	8.2	1.9	(32)
(33)		10%	WB	-0.4	1.1	2.2	6.6	10.2	13.6	14.5	14.7	11.9	7.9	5.0	0.2	(33)
(34)			MCDWB	0.7	3.1	4.6	11.1	14.1	18.7	19.4	19.7	15.5	11.6	6.7	1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.5	10.5	9.4	11.2	10.6	11.6	11.9	12.6	10.5	11.0	8.4	7.5	(35)
(36)			MCDBR	10.2	13.2	13.8	16.9	15.2	14.7	15.5	16.3	15.0	17.8	11.6	7.8	(36)
(37)			MCWBR	8.2	9.5	9.2	10.1	8.8	7.9	8.5	8.9	8.7	11.0	8.2	6.4	(37)
(38)		5% WB	MCDWR	8.9	10.5	12.5	16.5	14.0	14.1	13.9	14.2	13.6	16.2	10.1	6.8	(38)
(39)			MCWBR	7.4	8.1	8.4	9.9	8.5	7.9	7.9	8.2	8.4	10.2	7.3	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.273	0.300	0.337	0.390	0.376	0.374	0.388	0.401	0.362	0.329	0.304	0.276	(40)	
(41)		taud	2.381	2.314	2.299	2.212	2.306	2.338	2.296	2.258	2.353	2.451	2.464	2.423	(41)	
(42)		Ebn at Noon	844	880	891	867	891	893	874	847	854	839	799	802	(42)	
(43)		Edh at Noon	73	95	112	134	126	123	126	126	105	83	68	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.34	2.13	3.24	4.34	5.18	5.80	5.75	5.22	3.73	2.61	1.54	1.10	(44)	
(45)		RadStd	0.16	0.27	0.43	0.48	0.48	0.55	0.48	0.50	0.38	0.31	0.19	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (81 neighbors)	+0.64	N/A	N/A	+0.90	N/A	N/A	-117	-175	+93	+42	(47)

Nomenclature: See separate page